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MATHLOOTH

A Multimodal Pipeline for Arabic Text-to-2D-to-3D Saudi Cultural Object Generation

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Why do AI models fail to generate accurate Saudi cultural artifact ?

Introduction

Definition of the Problem

Heritage Geometry

Models show limited accuracy in representing Saudi cultural forms [13].

Arabic Semantics

Generative systems insufficiently capture Arabic linguistic meaning [14].

Cultural Motifs

Context-specific patterns lack precise visual reconstruction [15].

Dataset Bias

Western-centric datasets misrepresent Saudi cultural artifacts [6, 9].

Significance of the Study

Reduces the gap

In Saudi heritage representation [6][7].

Builds an Arabic-text 2D–3D

Generative pipeline.

Supports cultural preservation

In line with Vision 2030.

Builds Dataset

Provides a curated Saudi heritage dataset.

Introduces CAL

To enhance cultural fidelity.

Project Scope

1

Generation Pipeline

Develops a two-stage system for Arabic-to-2D and 2D-to-3D generation.

2

Cultural Focus

Targets authentic Saudi heritage objects for digital preservation.

3

Clear Boundaries

Includes only tangible artifacts; excludes animate figures and large scenes.



Related Work

Background

Significant progress in text-to-image and text-to-3D generation

using diffusion and transformer models [1, 2, 3, 4, 5].

Public datasets remain Western-centric, shaping the conceptual

and visual biases of generative models [6, 8, 9, 13, 27].

3D generative pipelines depend on linguistic, visual, and

geometric priors derived from large-scale training datasets [10, 11, 12].

Arabic and Saudi cultural data are underrepresented in current

datasets, limiting semantic and geometric accuracy [6, 9, 13, 14, 16].

Background Information

Model Type	Strengths	Limitations	Relevance to Our Work
VAEs [17]	Structured latent representation	Low detail / blurry	Not suitable for heritage details
GANs [18]	Sharp, realistic images	Unstable training / mode collapse	Not ideal for robust pipeline
Diffusion Models [1, 19]	Stable, high-quality, diverse	Slower than GANs	Chosen backbone for Text to Image

Similar Work Comparison

Stage	Models	Strengths	Limitations
Text to 2D	Stable Diffusion, DALL·E [3], [20].	Photorealistic images, strong generalization	No Arabic support, Western cultural bias, weak multi-object reasoning, missing Saudi visual cues
2D to 3D	TripoSR, TripoSG, Hunyuan3D [38] , RealFusion, Hi3DGen [21], [25], [23], [22].	Fast single-view 3D, good geometry priors	Lose fine ornaments, poor texture detail, no cultural semantics, trained on Western datasets
Text to 3D	HyperSDFusion, Michelangelo, Magic3D, DreamFusion [4], [24], [29], [28].	Strong structure learning, improved semantic–geometry mapping	No Arabic embeddings, lacks cultural fidelity, fails with symbolic heritage shapes

Commercial Tools

Tool	Input	Language Support	Cultural Diversity
Genie (Luma AI), [30]	Text	English-only	×
Scenario, [31]	Text / Image	Partial Multilingual	×
Meshy AI, [32]	Text / Image	English-only	×
Ours	Text/Image	Arabic/English	✓

Methodology





Dataset

Why did we construct our own data?

1

Lack of culturally specific contexts

Existing datasets do not include Saudi heritage objects or their cultural meanings.

2

Missing Multimodal Alignment

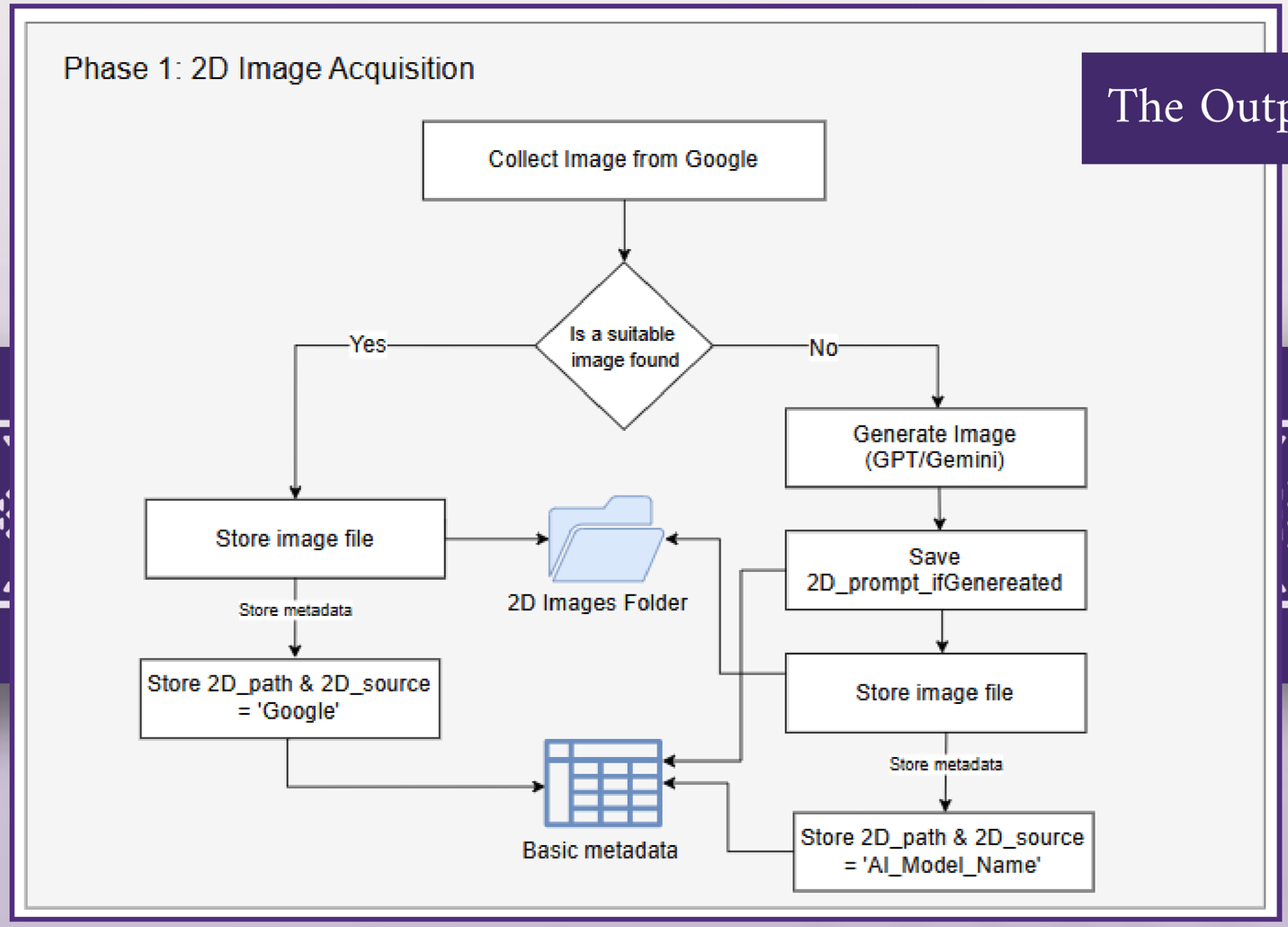
Available datasets do not provide paired text, 2D images, and 3D models for the same item.

3

Lack of Saudi Cultural Semantics

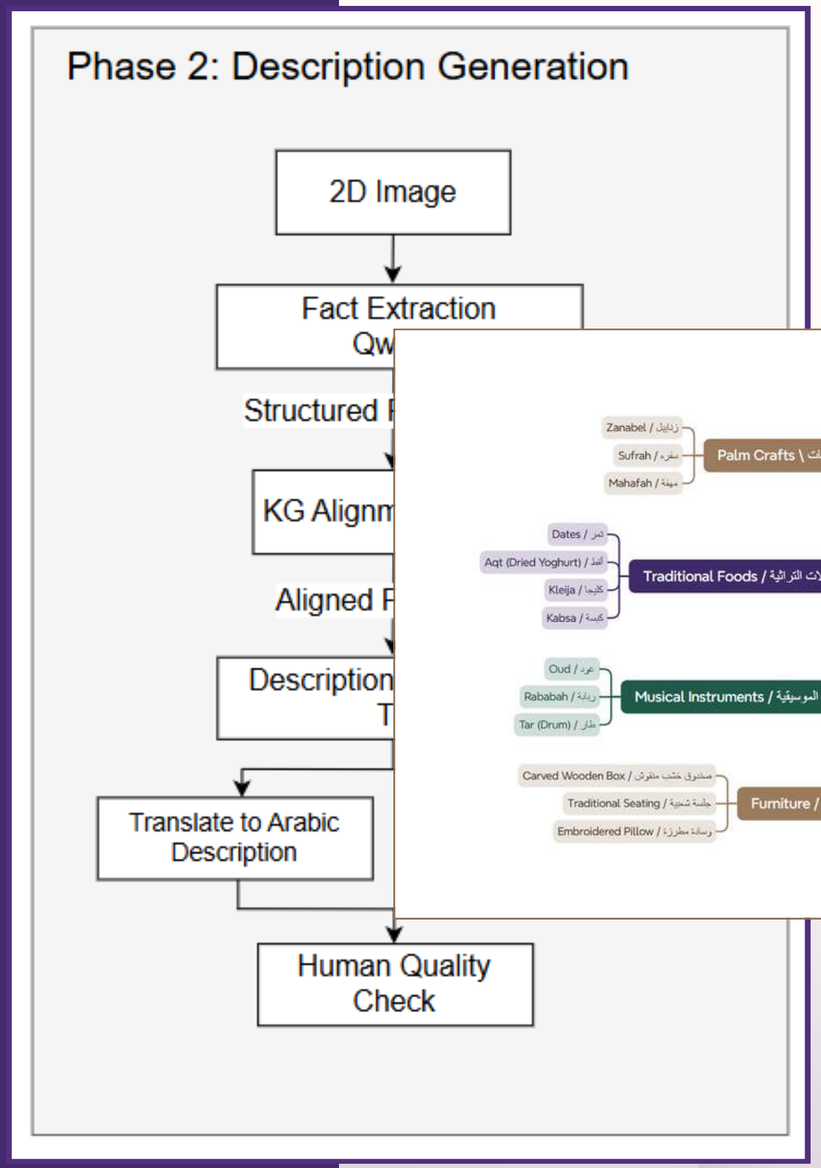
Existing datasets do not capture object parts, motifs, attributes, and meanings needed for culturally grounded generation.

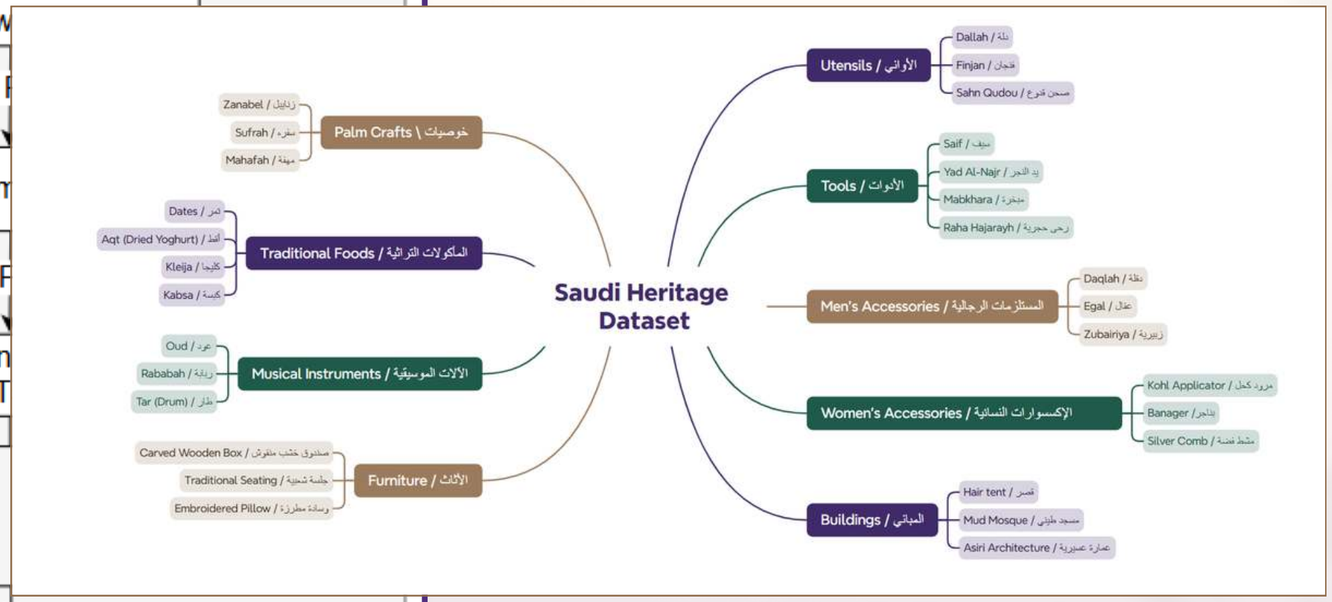
Phase 1



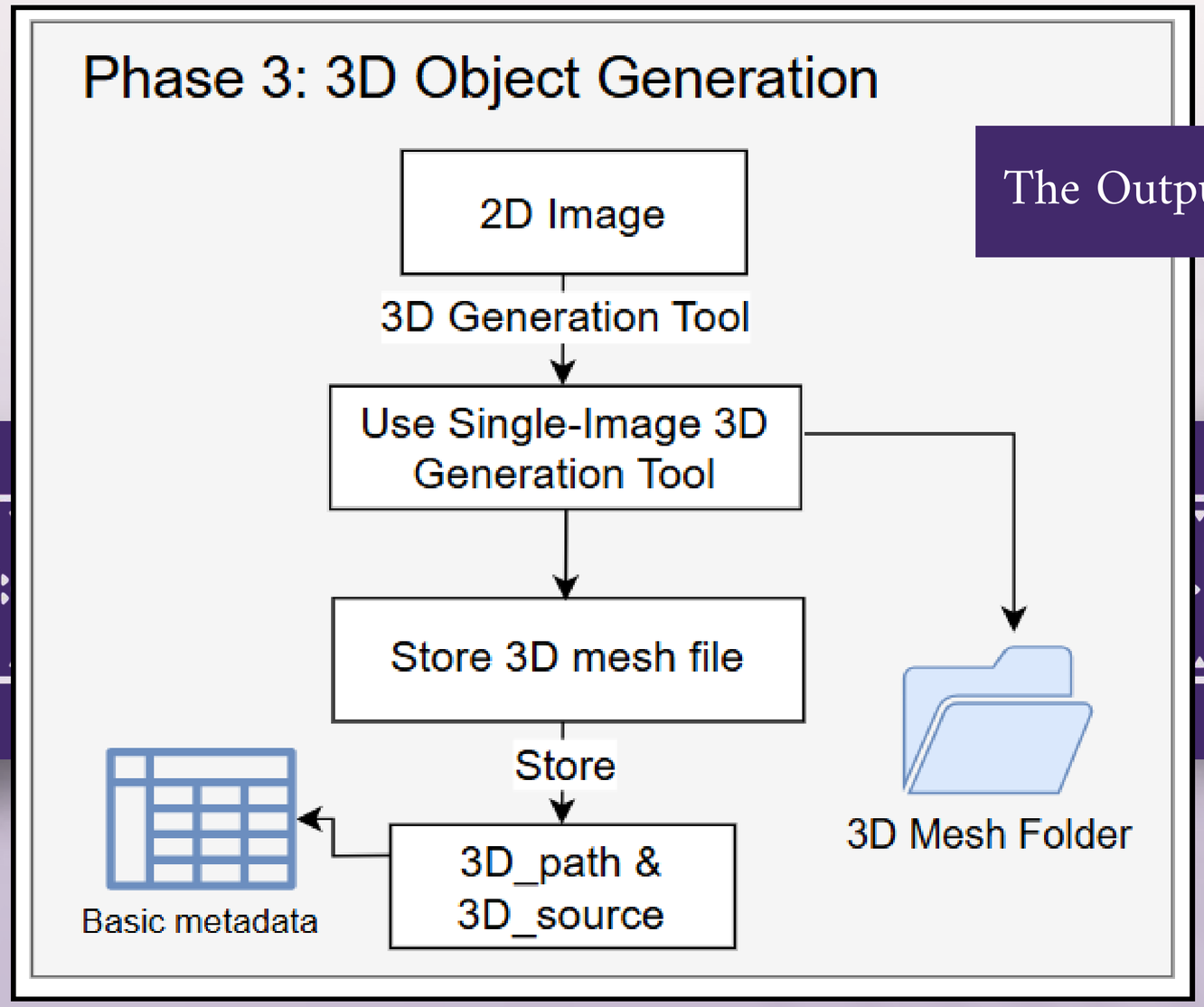
The Output: 2D images folder

Phase 2


Phase 2: Description Generation
2D Image
Fact Extraction
Qw
Structured
KG Alignm
Aligned
Description
T
Translate to Arabic Description
Human Quality Check


Saudi Heritage Dataset
Palm Crafts \ خوصيات
Zanabel / زنابيل
Sufrah / سفره
Mahafah / مهفة
Traditional Foods / المأكولات التراثية
Dates / تمر
Aqt (Dried Yoghurt) / أقط
Kleija / كليجا
Kabsa / كبسة
Musical Instruments / الآلات الموسيقية
Oud / عود
Rababah / ربابة
Tar (Drum) / طار
Furniture / الأثاث
Carved Wooden Box / صندوق خشب منقوش
Traditional Seating / جلسة شعبية
Embroidered Pillow / وسادة مطرزة
Utensils / الأواني
Dallah / دلة
Finjan / فنجان
Sahn Qudou / صحن قدوع
Tools / الأدوات
Saif / سيف
Yad Al-Najr / يد النجر
Mabkhara / مبخرة
Raha Hajarayh / رحى حجرية
Men's Accessories / المستلزمات الرجالية
Daqlah / دقلة
Egal / عقال
Zubairiya / زبيرية
Women's Accessories / الإكسسوارات النسائية
Kohl Applicator / مرود كحل
Banager / بناجر
Silver Comb / مشط فضة
Buildings / المباني
Hair tent / قصر
Mud Mosque / مسجد طيني
Asiri Architecture / عمارة عسيرية

Phase 3



The Output: 3D Meshes folder

Dataset Categories and Subcategories

Multi-Modality Dataset Preprocessing

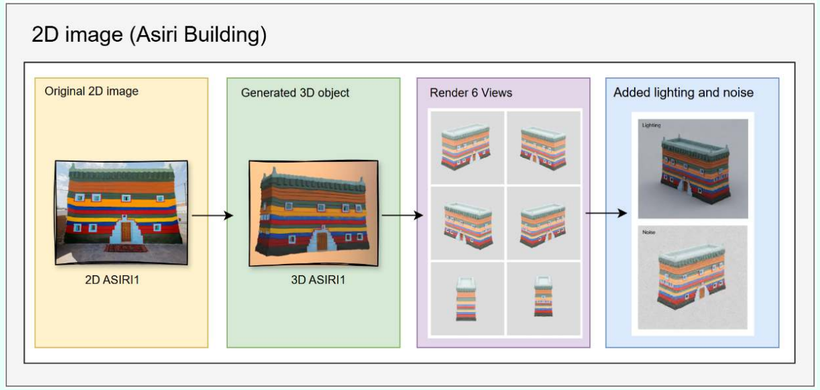
Modality	Preprocessing
Text	Normalize, Tokenize
2D Images	Remove background, Fix lighting , Standardize size
3D Meshes	Validate geometry , Clean mesh

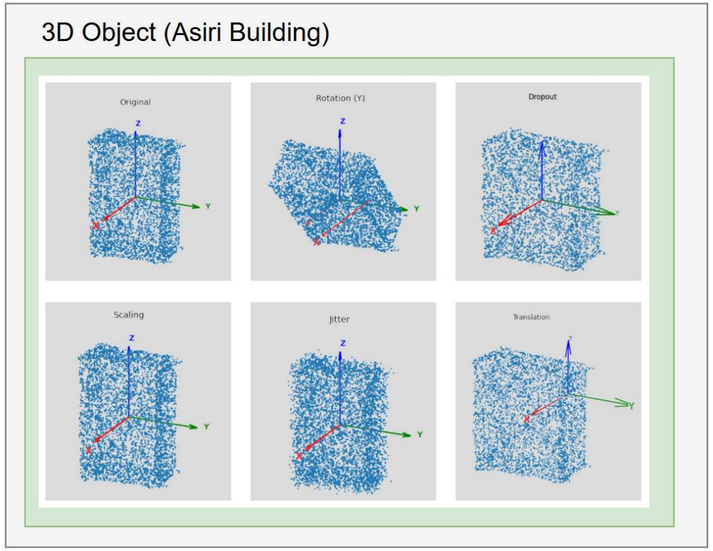
Multi-Modality Dataset Augmentation

Text	2D images	3D objects

Text (Asiri Building)

Sample ID	Description
ASIRI1	Asiri Architecture: Rectangular stone structures with stepped exteriors, repeating windows, symmetrical facades, and colorful matte-plastered surfaces decorated with geometric designs.
ASIRI2	Tall, multistory rectangles with stepped facades, mattesized brown-and-white plastered walls featuring geometrical Qatt designs.
ASIRI3	Rectangular, multi-storied structures with stepped design, stone-mud façade in vibrant hues, featuring symmetrical layouts and minimal windows.
ASIRI4	Asiri Architecture: Rectangular skyscrapers with stepped massing, tall windows, repeated upper-floor designs, symmetrical facades covered in stone and whitewashed geometric patterns.





From ~300 samples to ~5000 samples

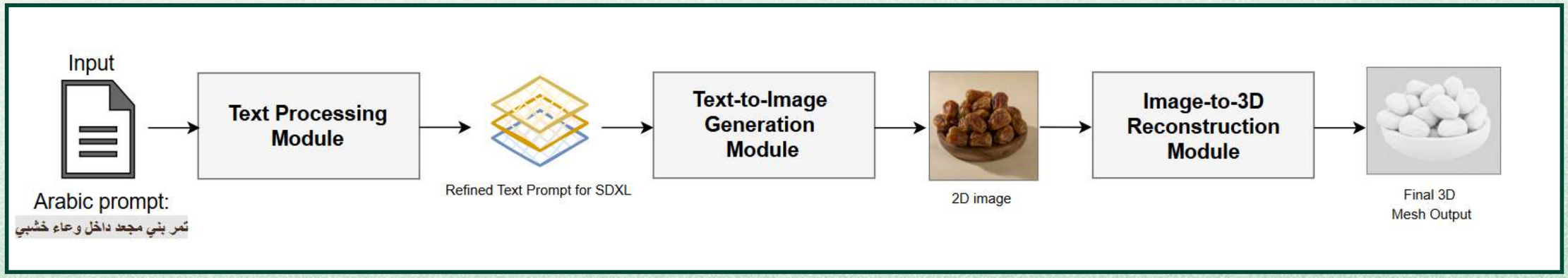
Sample of 3 modality Dataset

Main Category	Subcategory(s)	Baseline Description	Rewritten Description	2D Image	3D Image
Furniture	Sanduq (صندوق)	A rectangular chest with a flat or slightly domed lid, where width significantly exceeds height. The lid is thicker than the side panels, and the structure rests on small hidden feet. The front face exhibits bilateral symmetry and includes badan, ghita, maqbad, mafasil, and zakhrafa elements. The surface is matte or semi-matte with hand-painted geometric motifs in brown, red, blue, and white, often complemented by iron hardware.	Traditional Sanduq chest, rectangular box form with flat or slightly domed lid. Width larger than height, thick lid, bilateral symmetry. Includes badan, ghita, maqbad, mafasil, zakhrafa. Matte painted surface in brown, red, blue, white with geometric motifs and iron hardware.		

Samples of 3D Data

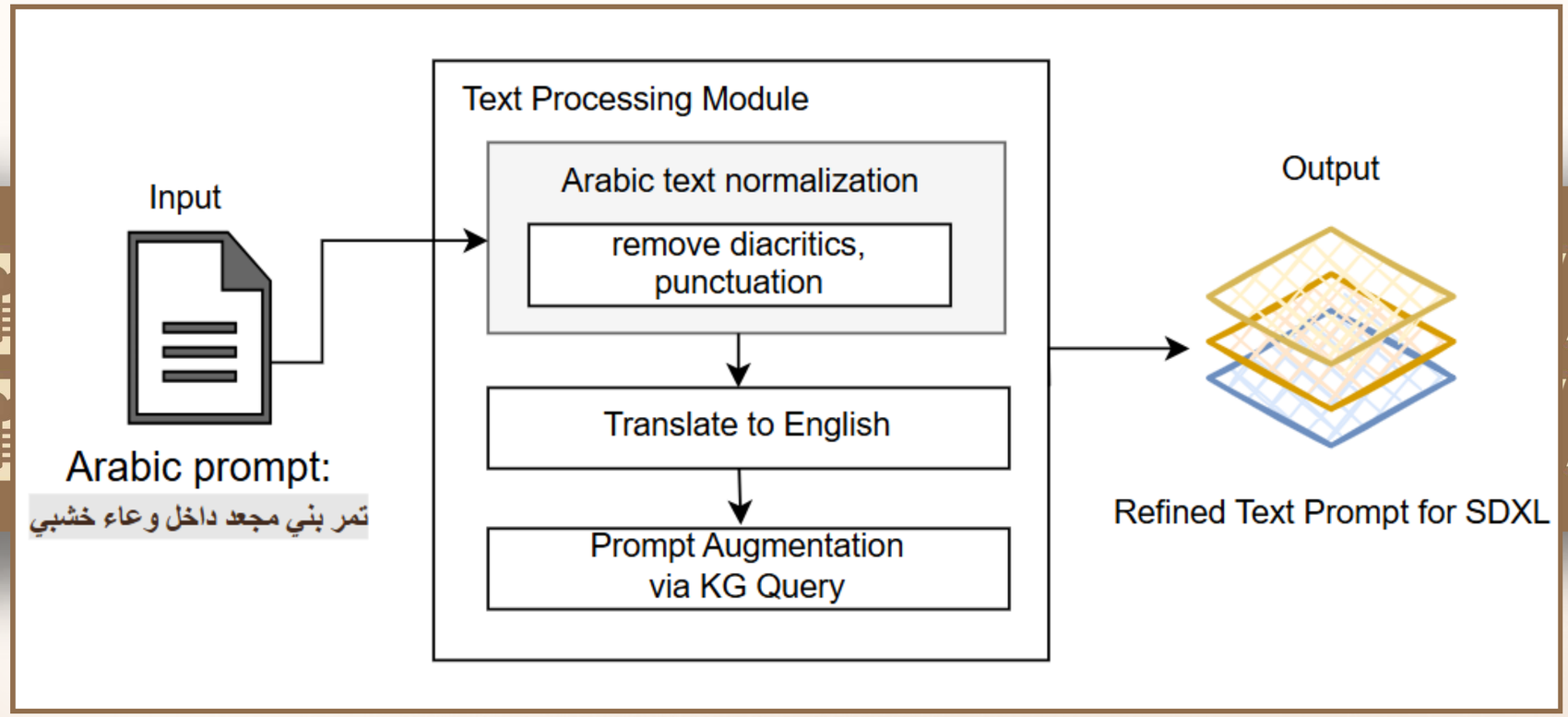


Proposed Solution



High-level overview of the proposed Text-to-2D-to-3D generative pipeline

Text Processing Module



How the KG Refines the Prompt

A neuro-symbolic layer that adds cultural facts before image generation

1 **Raw Arabic Prompt**

تمر بني مجعد داخل وعاء خشبي

2 **KG Facts & Constraints**

- Object: Dates
- Material: Wooden bowl
- Texture: natural wrinkles
- Avoid: plastic / modern container

3 **Enhanced Prompt**

A traditional Saudi-style wooden bowl filled with wrinkled brown dates, natural texture, culturally accurate serving object, realistic lighting.

Output: A culturally grounded prompt sent to fine-tuned SDXL

Text-to-Image Generation Module

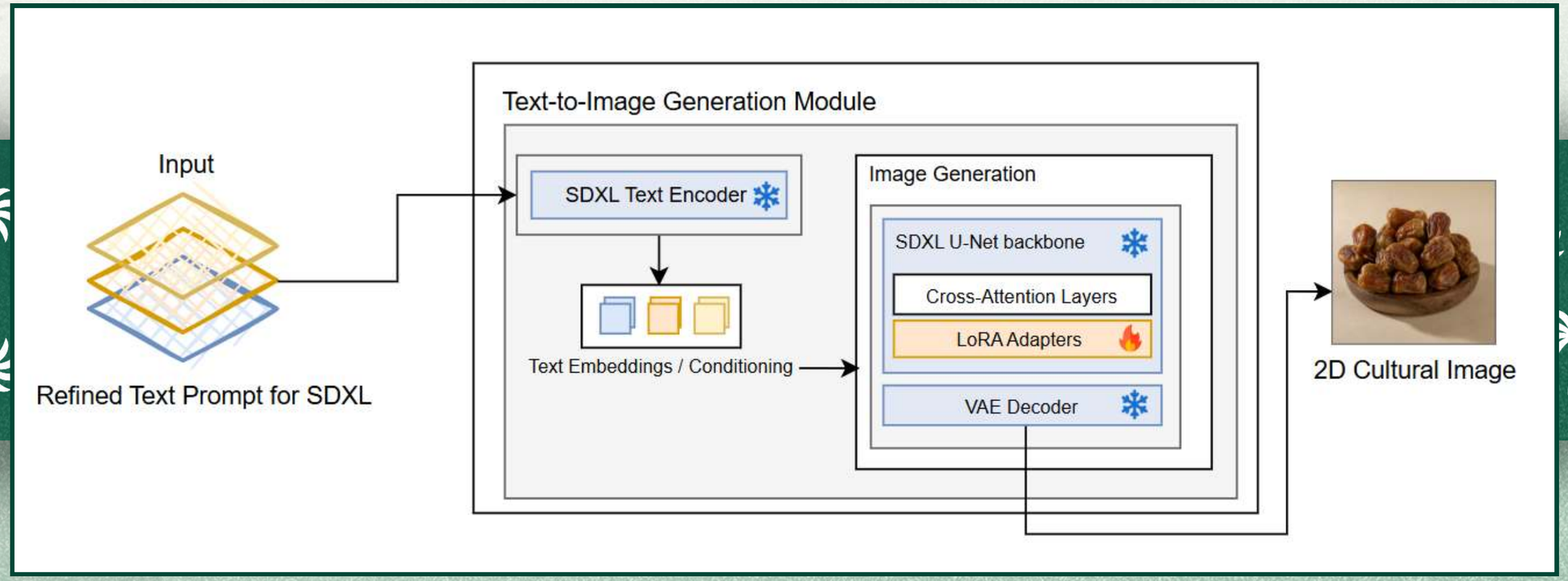
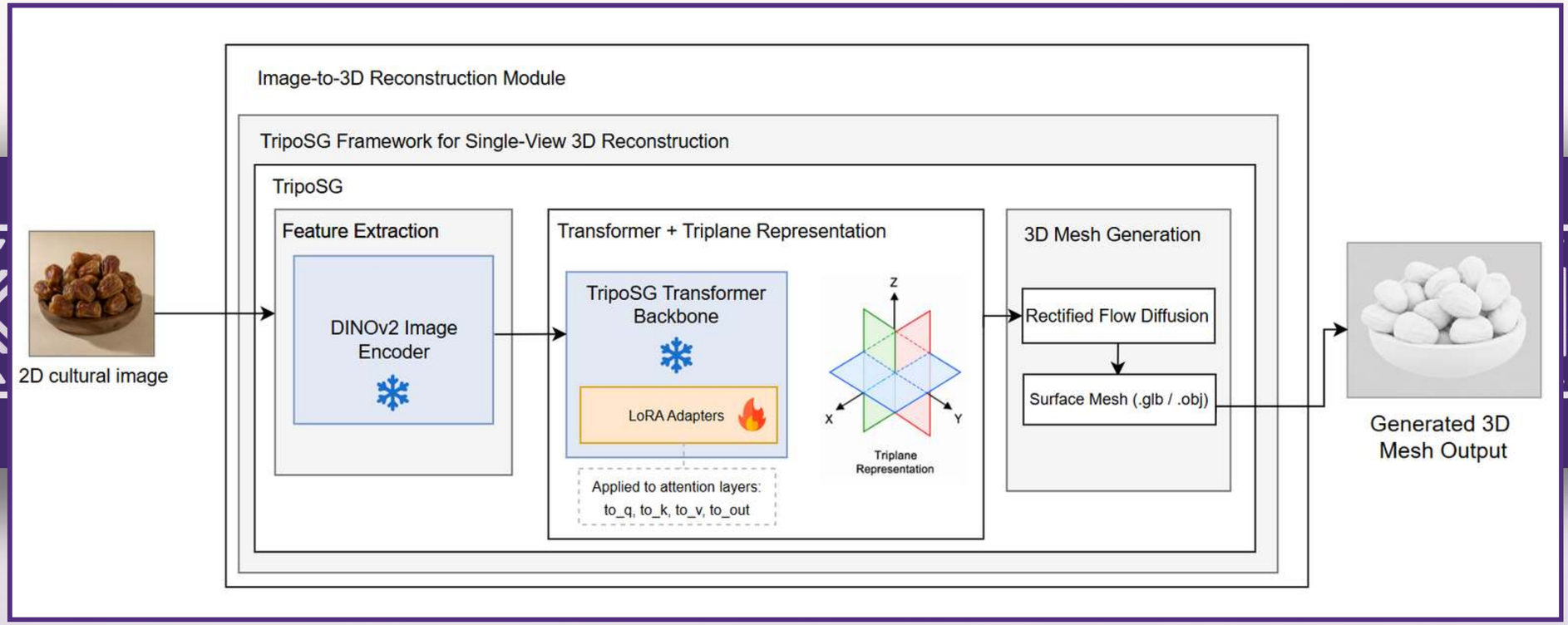
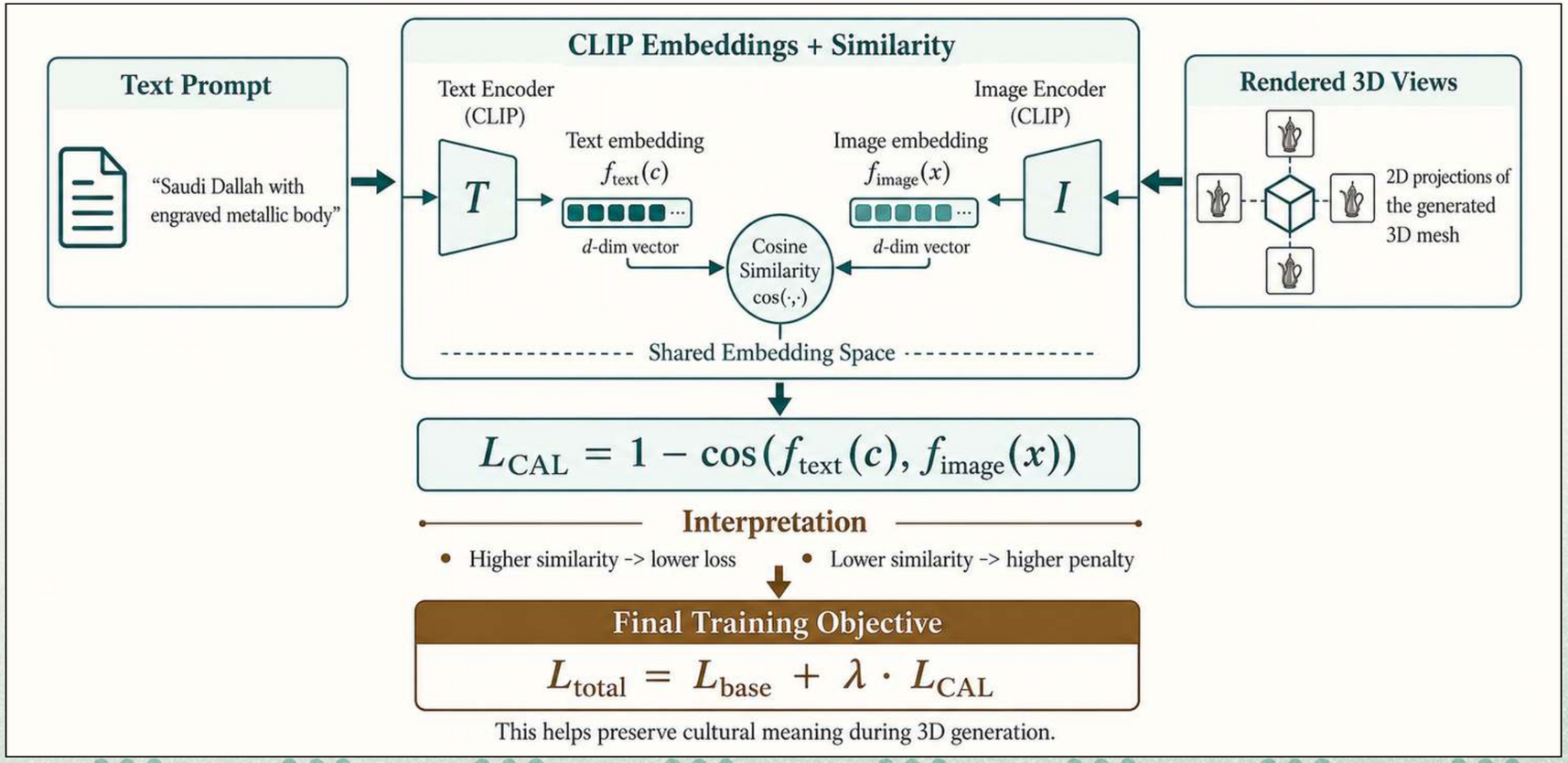


Image to 3D Reconstruction Module

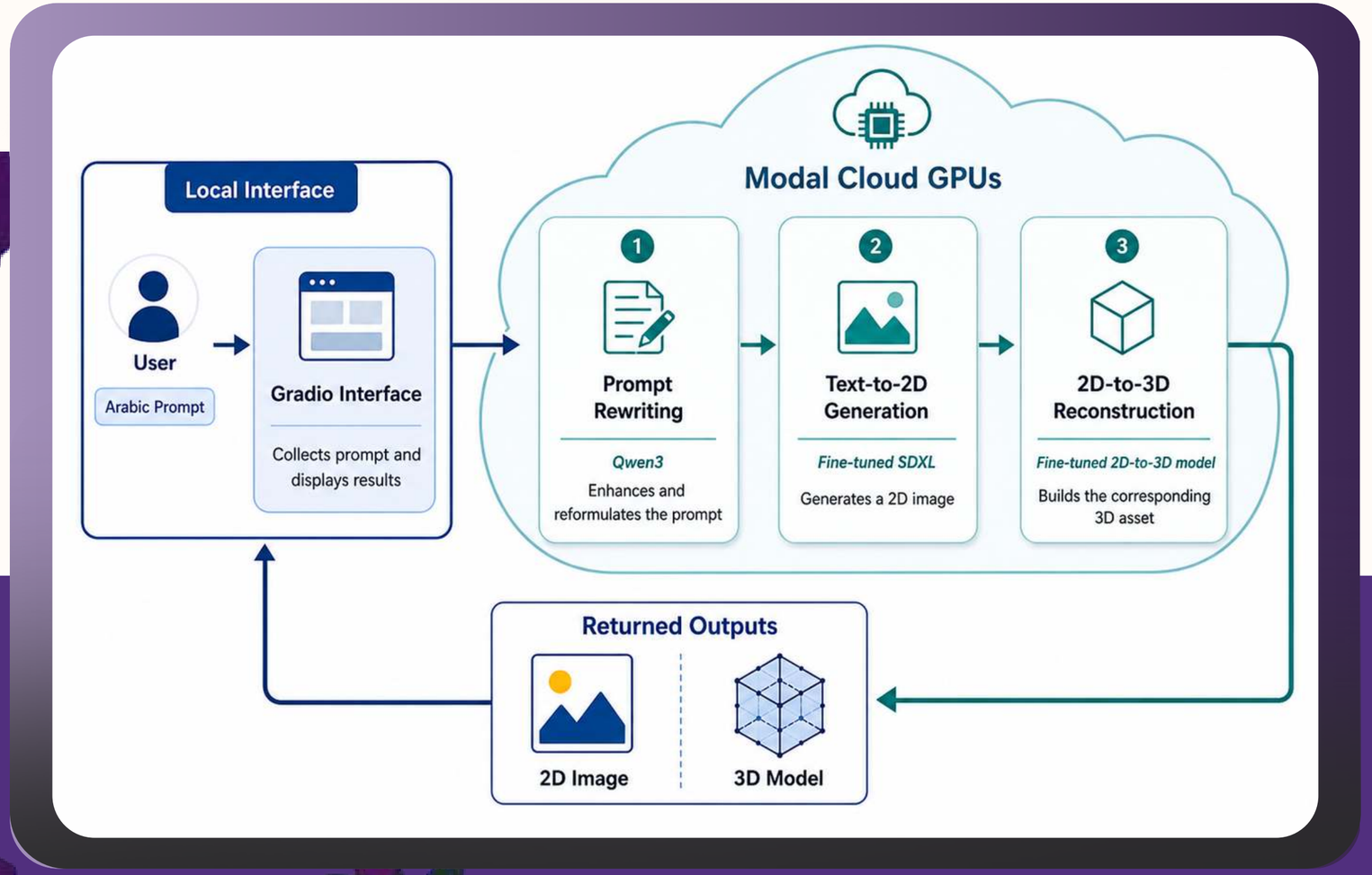


Cultural Alignment Loss

Encourages semantic consistency between the text prompt and the generated 3D object



High-level integrated inference workflow



Results and Evaluation

Evaluation Metrics

Quantitative Metrics

Text-to-2D Evaluation

- FID, KID
- CLIP Score, Long-CLIP Score

Image-to-3D Evaluation

- Chamfer Distance
- CLIP Score, CAL Similarity Score
- Noise Prediction MSE

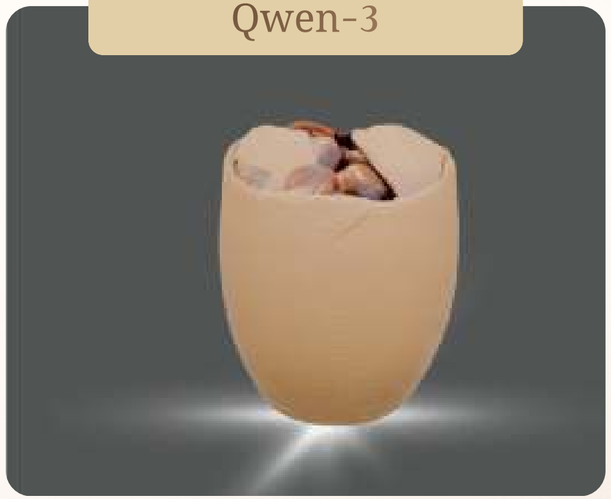
Text-to-2d Phase

Stable Diffusion XL - test sample

Baseline Captions

“A bowl containing dates”

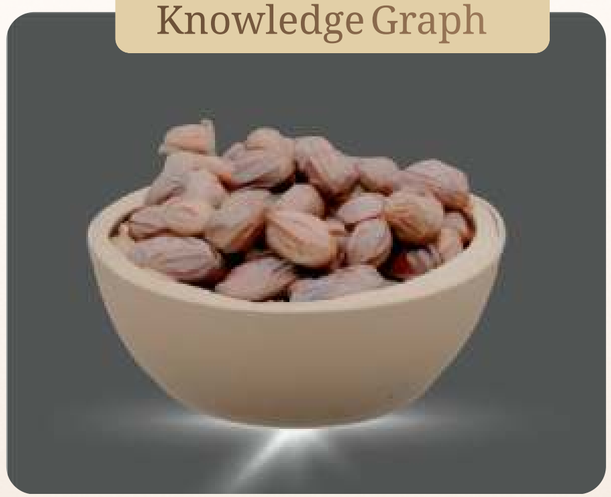
Qwen-3



Rewritten Captions

“Dates with oval elongated shapes, slightly wrinkled brown skin, and natural color variations from dark brown to amber, placed inside a wooden bowl”

Knowledge Graph



Text-to-2d Phase

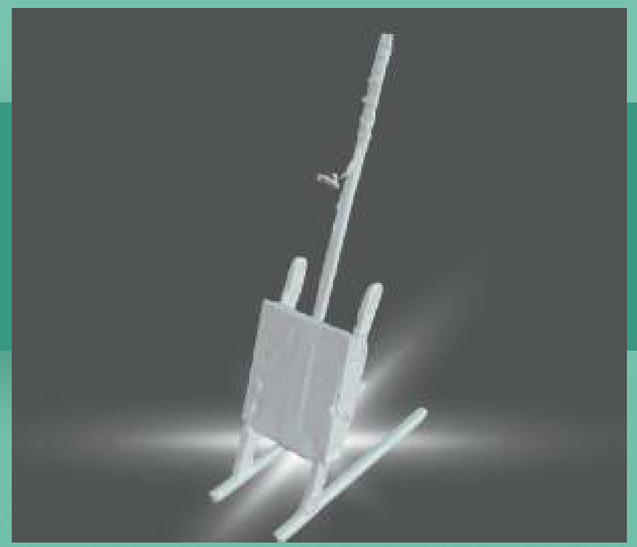
Fine- tuned Stable Diffusion XL Results

Metric	Fined-tuned SDXL Baseline Captions	Fined-tuned SDXL Rewritten Captions
FID(↓)	223.31	**174.86**
KID(↓)	0.07984	**0.02641**
CLIP(↑)	0.26046	**0.26267**
Long-CLIP(↑)	0.26062	**0.27373**



2D-to-3D Phase

Fined-tuned TripoSG without CAL



Fined-tuned TripoSG with CAL

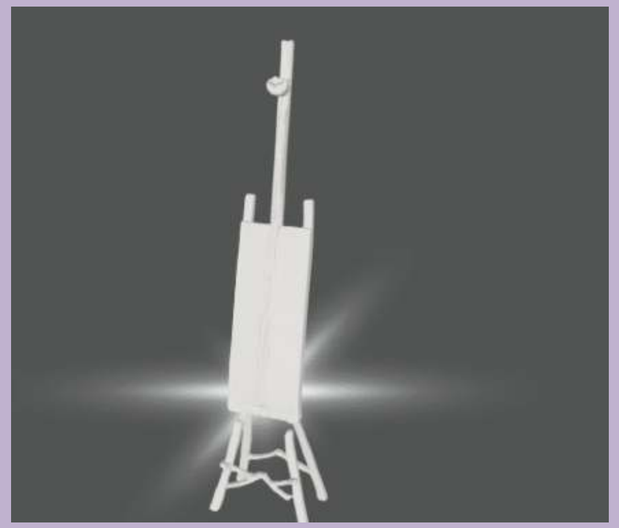


TripoSG Fine-tuning Results

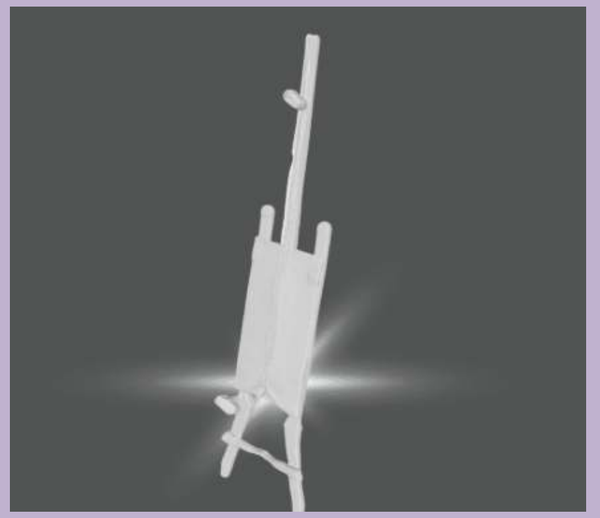
Model Configuration	Noise Pred MSE (↓)	Chamfer Distance (↓)	CLIP Score (↑)	CAL Score (↓)
Fined-tuned TripoSG Without CAL	**1.952**	**0.1972**	0.8165	-
Fined-tuned TripoSG With CAL	2.2704	0.2126	**0.8309**	**-0.2155**

2D-to-3D Phase Cross-Model Validation

Hunyuan3D - test sample



Fine-tuned with CAL



Fine-tuned without CAL

Fine- tuned Hunyuan3D Results

Model Configuration	Chamfer Distance (↓)	CLIP Score (↑)	CAL Score (↓)
Fine-tuned Hunyuan3D without CAL	0.171447	0.2296	-
Fine-tuned Hunyuan3D with CAL	**0.16683**	**0.2311**	-0.2311



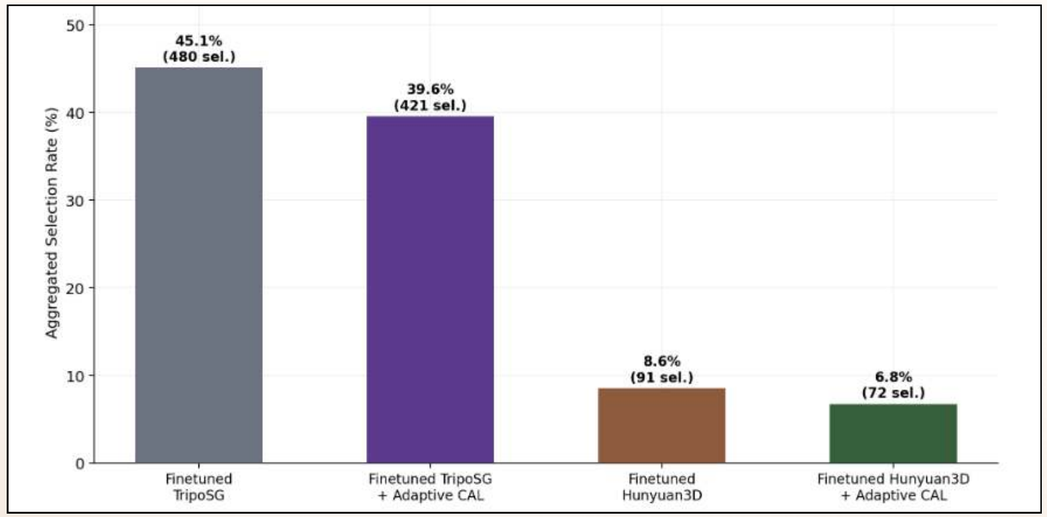
Evaluation Metrics

User-Based Evaluation

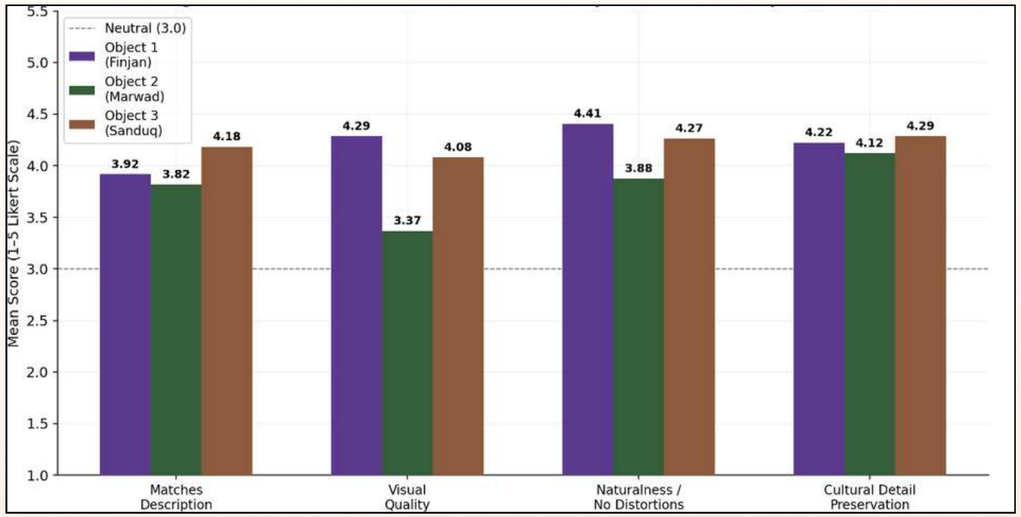
- Matches Description
- Visual Quality
- Cultural Detail Preservation
- Geometry Correctness
- Surface Quality
- Smoothness

Evaluation Metrics

User-Based Evaluation



Aggregated four-model comparison across all evaluated objects and criteria



Text-to-2D evaluation mean scores per criterion and object.

Findings

Findings

KG-guided captions

Improved Text-to-2D image quality and semantic alignment.

Rewritten captions

With fine-tuned SDXL achieved better FID, KID, CLIP, and Long-CLIP results.

Excessive epochs

Caused unstable 3D mesh generation.

Object-level splitting

Prevented data leakage between rendered views of the same object.

Increasing Data Size

Excessive scaling introduced noisy and fragmented 3D outputs.



Future Work

Future Work

Texture Refinement Module

Improve materials, colors, and decorative details of generated 3D objects.

Arabic Cultural Evaluation

Build a culturally aware Arabic vision-language evaluation model.

Expert Cultural Validation

Evaluate outputs with Saudi heritage specialists for better cultural accuracy.

Layered Cultural Alignment

Apply alignment at global, object, and attribute levels.

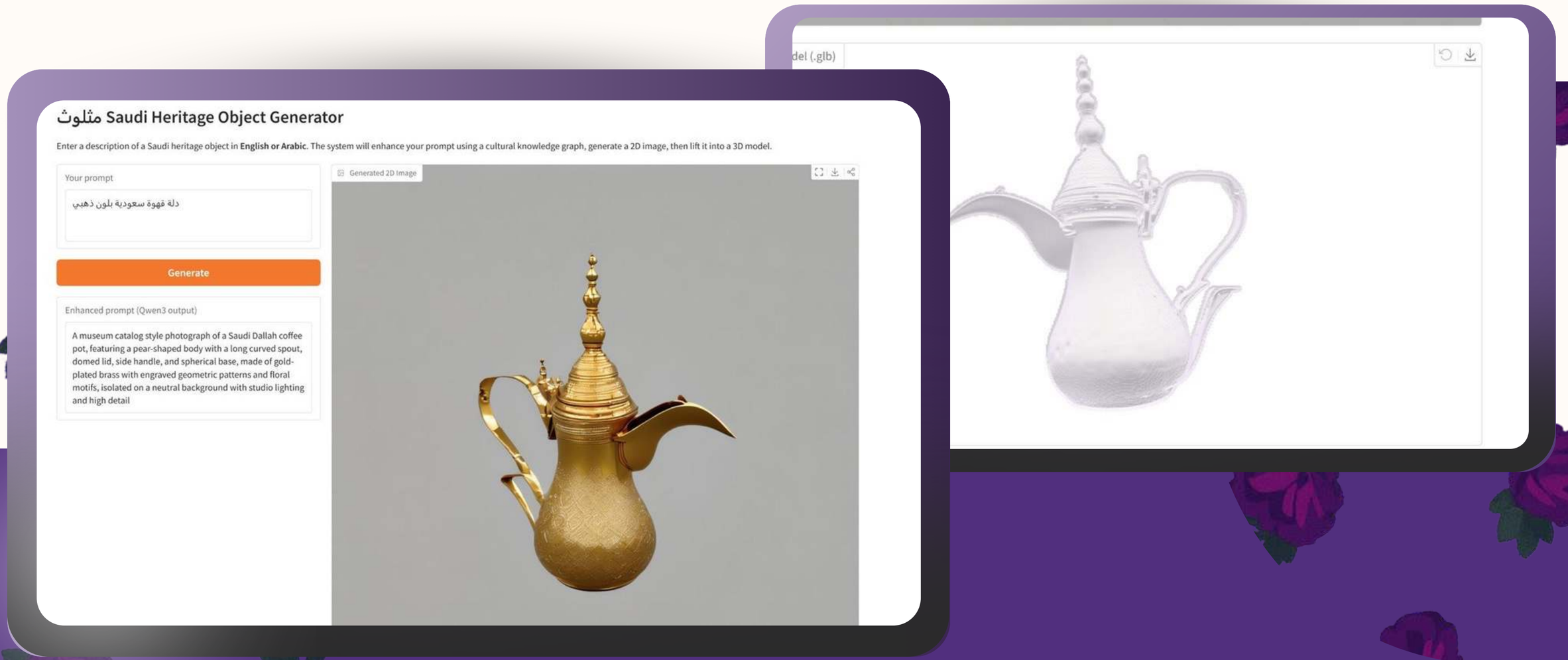
Multi-View Reconstruction

Use multiple object views to improve 3D reconstruction quality.

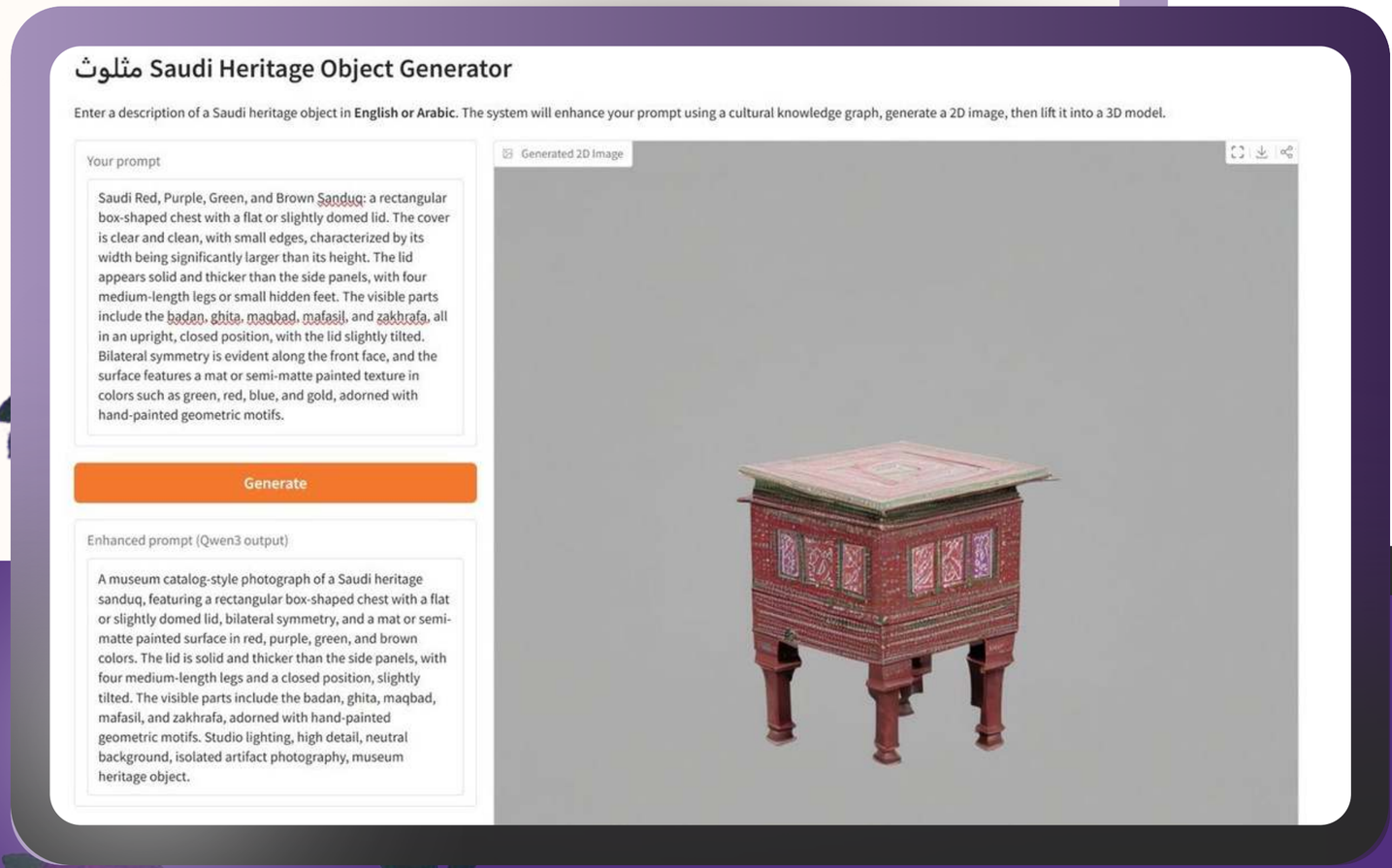
Dataset Expansion

Add more Saudi regions, categories, and heritage objects.

Demo from text to 3D



Demo from text to 3D





Conclusion





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من عمق التراث... إلى آفاق المستقبل

Thank You

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